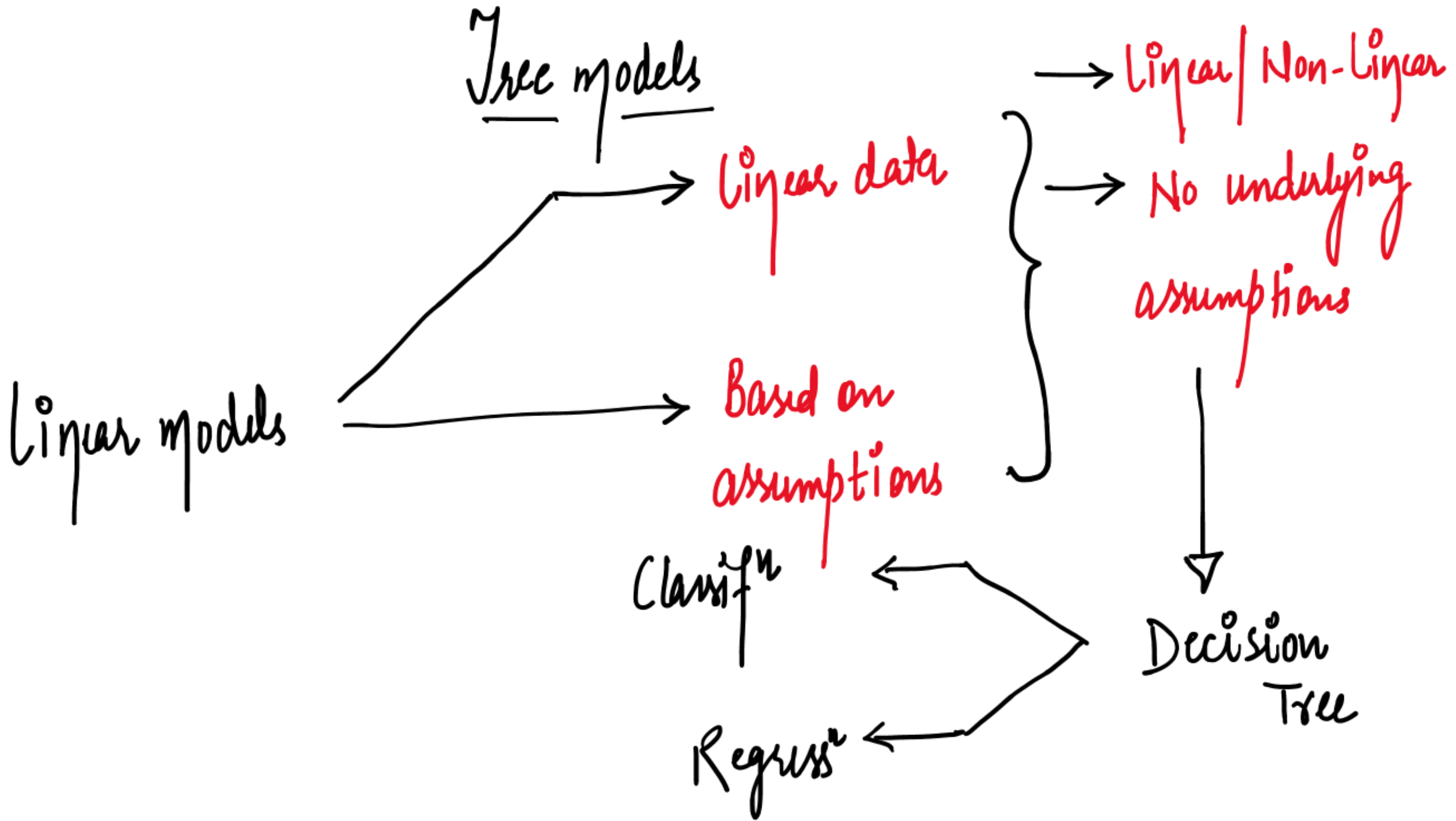
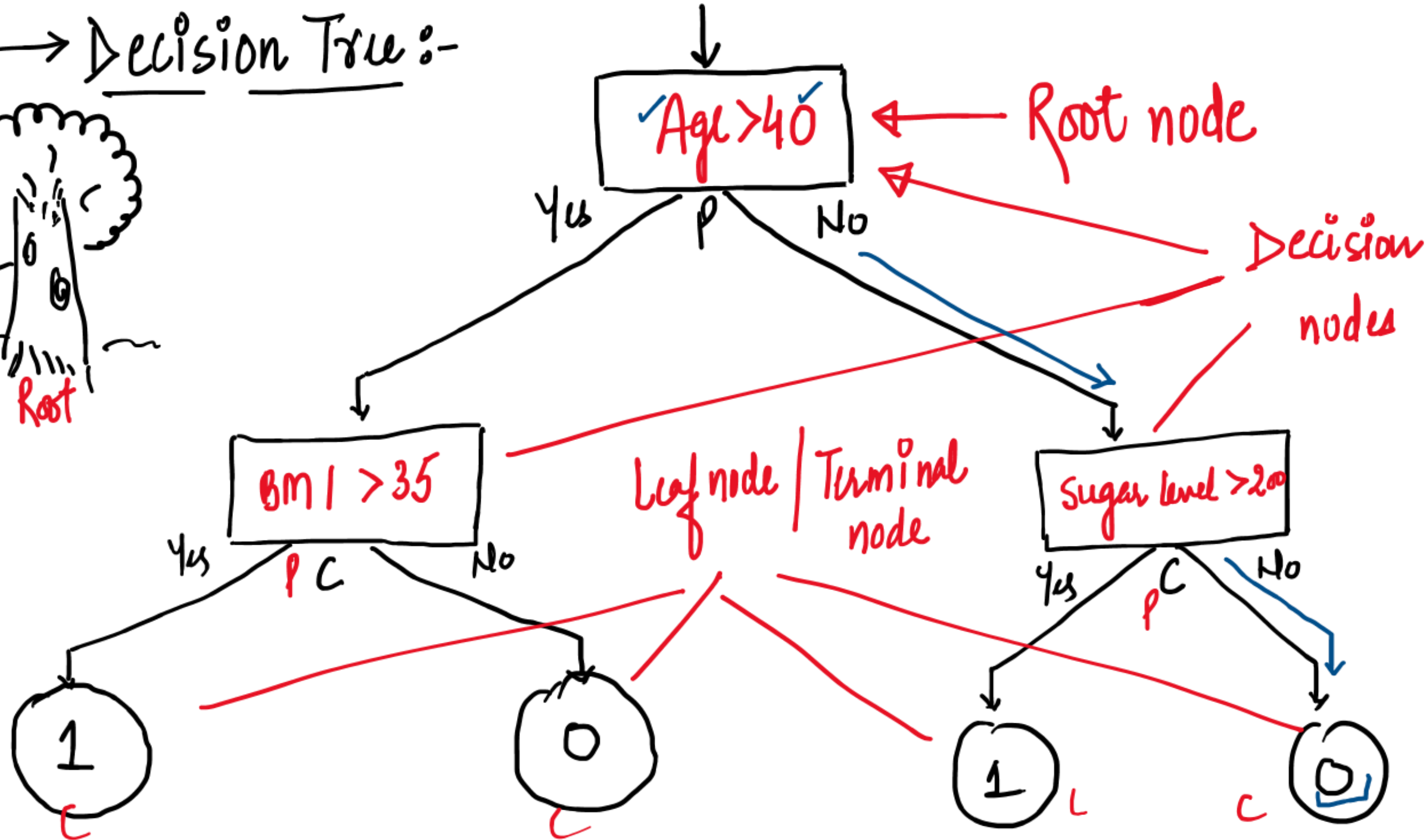




→ Problem:- Predict whether a patient is diabetic or not basis age, bmi, sugar level, gender

→ 28, 37, 180 → Not Diabetic

→ Decision Tree :-

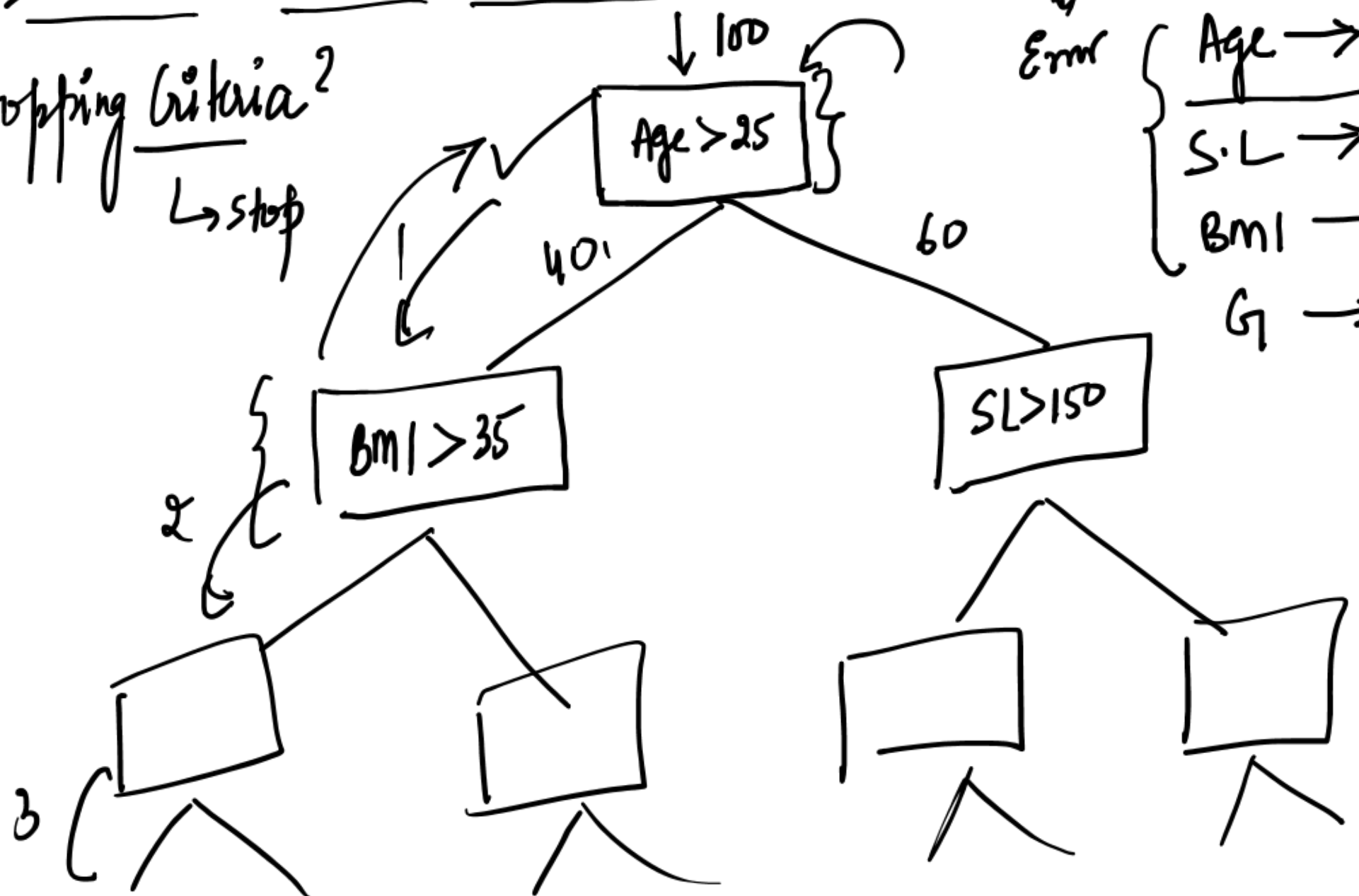


→ Decision Tree Construction :- ID3 / CHAID ✓

→ Stopping Criteria?
↳ stop

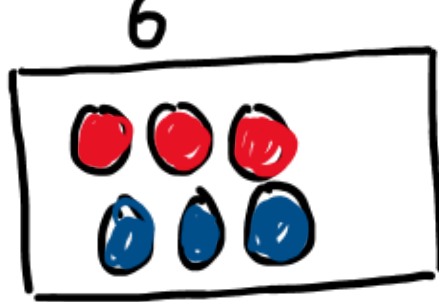
Err

$\left\{ \begin{array}{l} \text{Age} \rightarrow 25 \checkmark \text{E} \downarrow \\ \text{S.L} \rightarrow 150 \checkmark \\ \text{BMI} \rightarrow 30 \checkmark \\ \text{G} \rightarrow \text{M} \checkmark \end{array} \right\}$

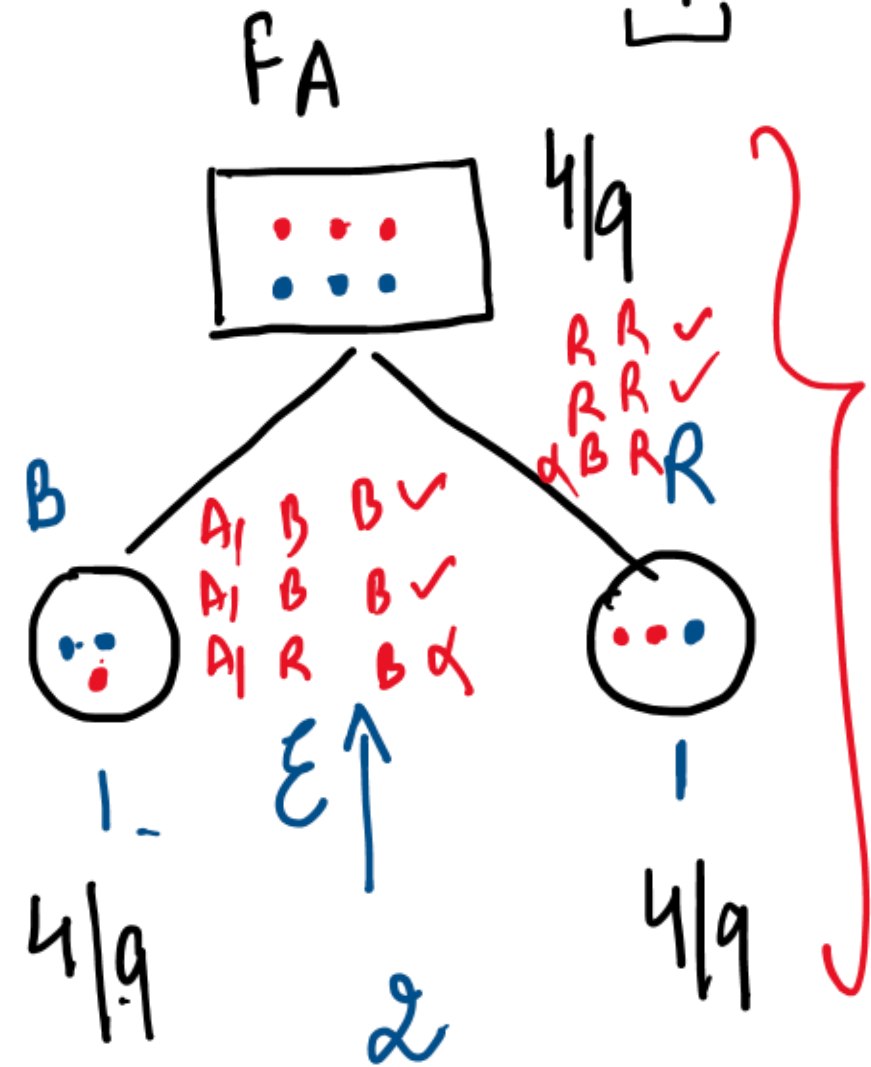


$$G_B = 1 - \sum p_i^2$$

$$= 1 - \left(\left(\frac{2}{3} \right)^2 + \left(\frac{1}{3} \right)^2 \right) = \frac{4}{9}$$

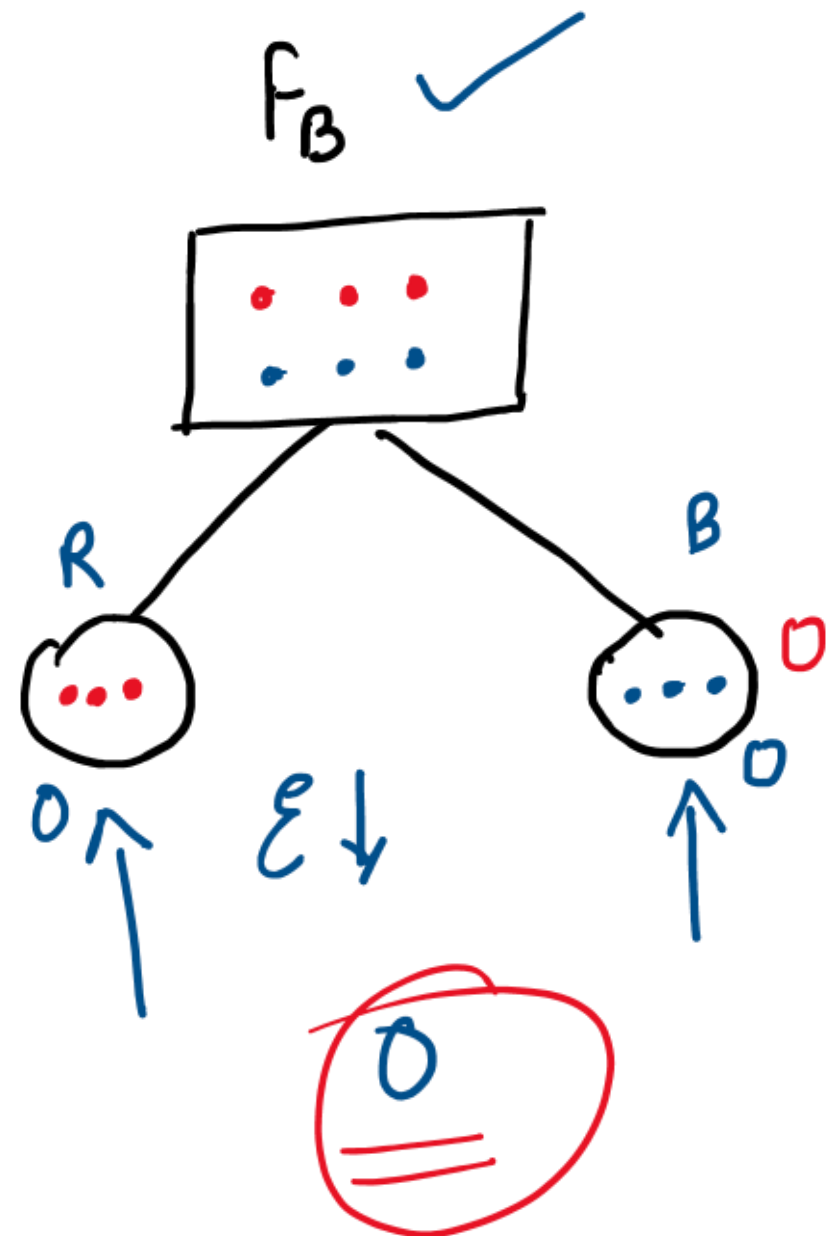


$$1 - \left[\left(\frac{2}{3} \right)^2 + 0 \right] = 0$$



$$\epsilon_A > \boxed{\epsilon_B}$$

$G_A > G_B$ ✓
 ✓ 0
 4/9

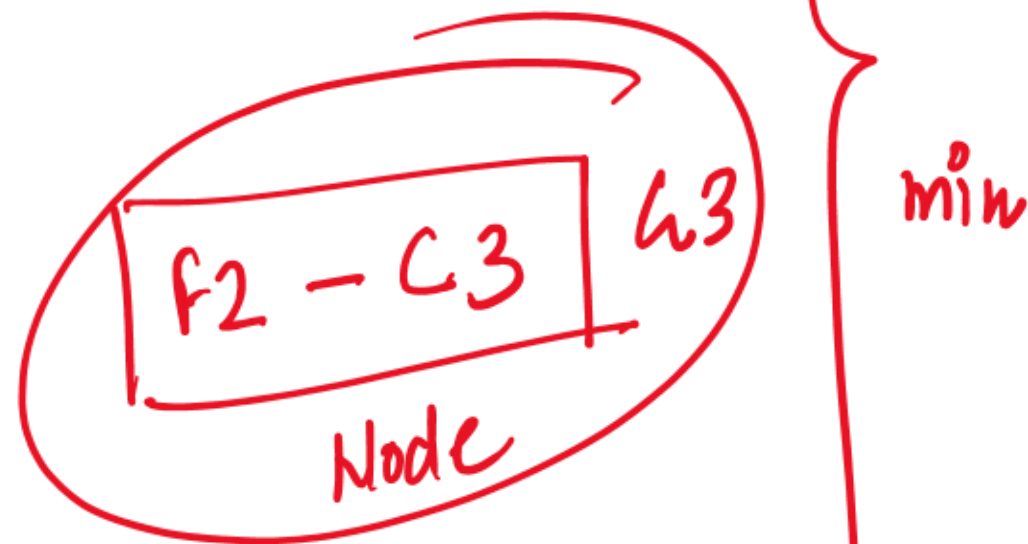
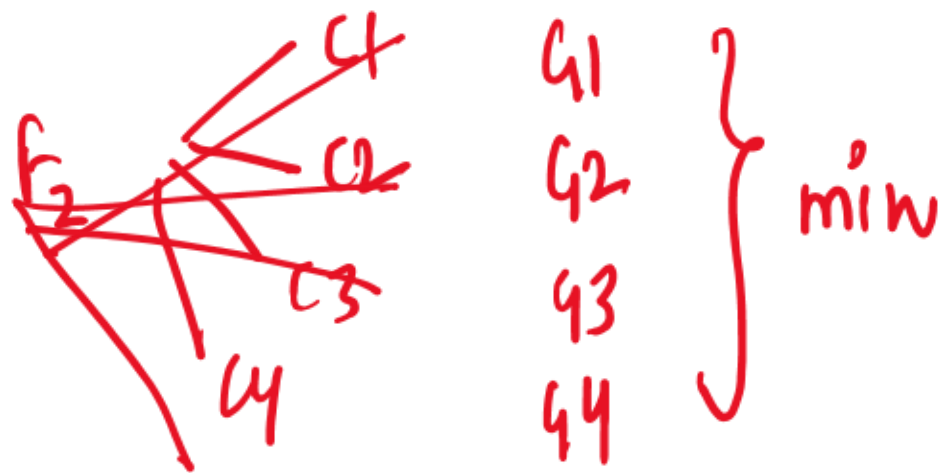
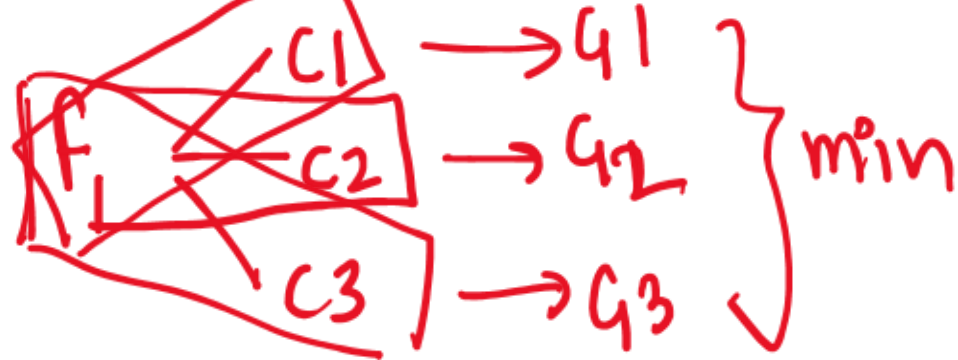


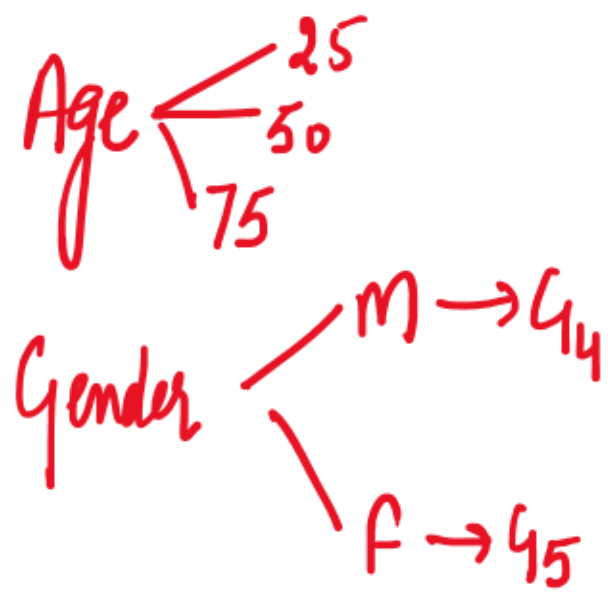
NI  $p_r = \frac{2}{3}$ $p_b = \frac{1}{3}$

$$G_N = 1 - \sum (p_r^2 + p_b^2) = 1 - \left[\left(\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 \right]$$

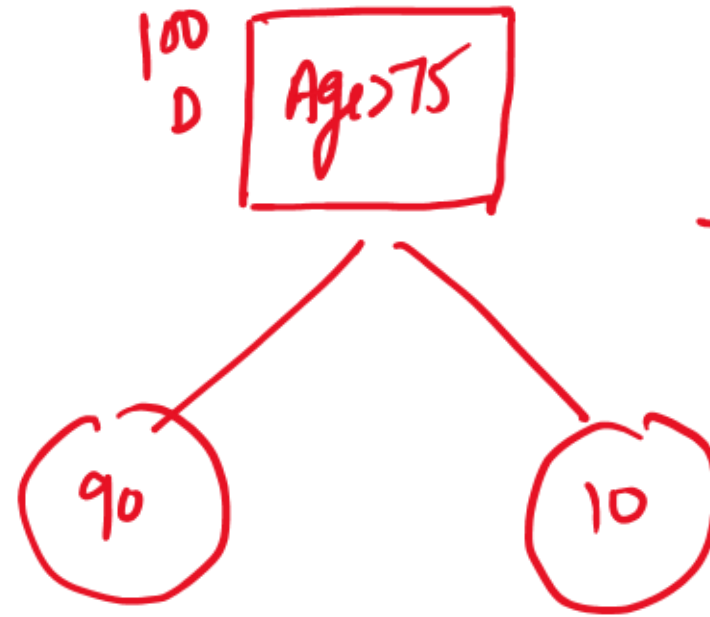
$$= 1 - \left[\frac{4}{9} + \frac{1}{9} \right] = 1 - \frac{5}{9} = \frac{4}{9}$$

$$G_S = \frac{G_{N1} + G_{N2}}{2} = \frac{\frac{4}{9} + \frac{4}{9}}{2} = \frac{4}{9}$$





Age > 75



→ G₁

G₁
G₂
G₃

Node

Age > 25

Age > 50

Age > 75

G = m

h = f

G₁

G₂

G₃

G₄

G₅

→ min

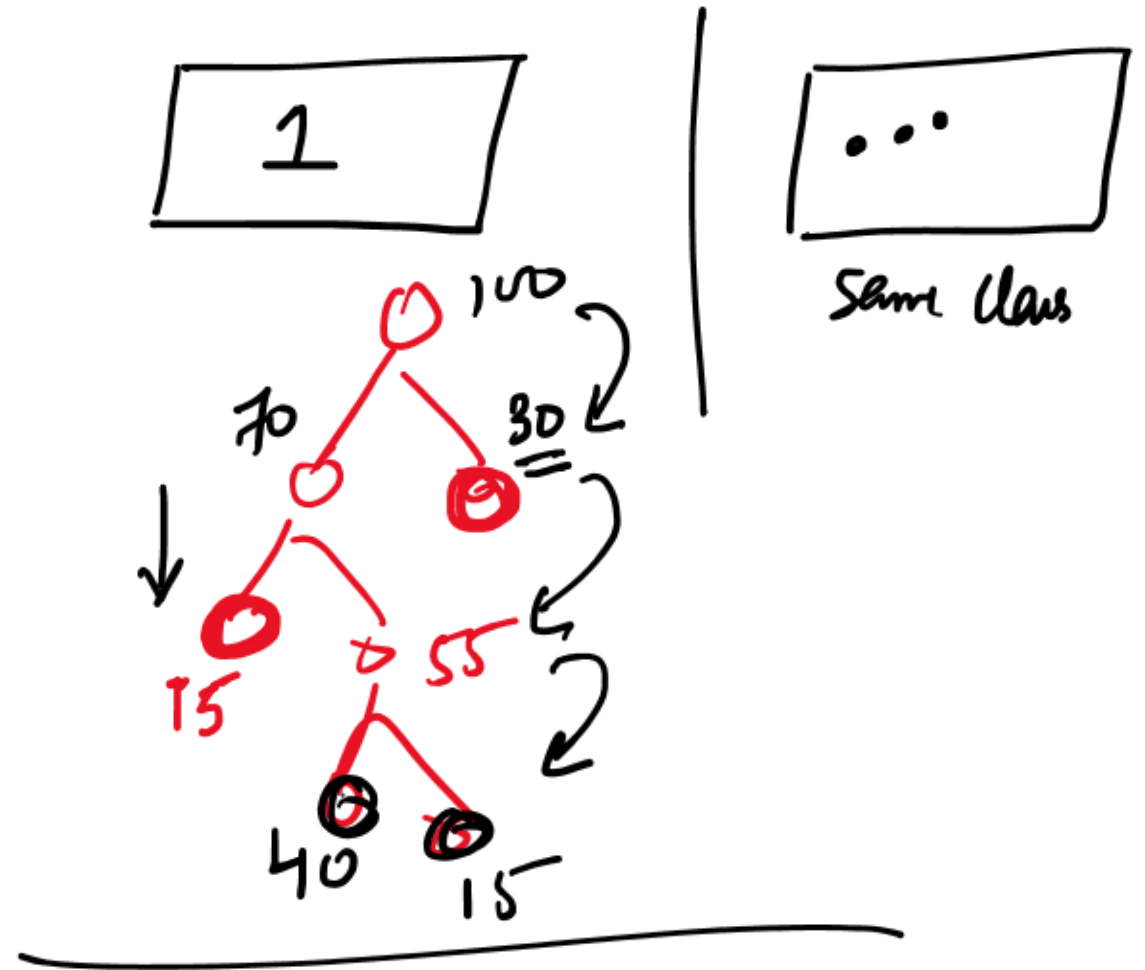
Gini $\rightarrow$ Statistical representⁿ of misclassifⁿ

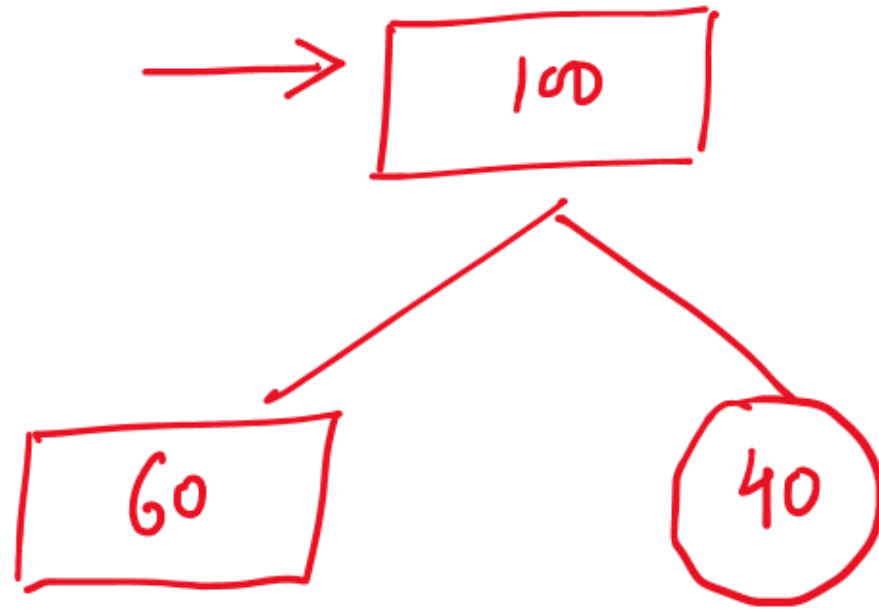
$$G = 1 - \sum p_i^2$$

$m \uparrow$ $G \uparrow$

→ Hyperparameters:

- ① max_depth :- Tree depth 3
- ② min_samples_split : 50
- ③ Criterion : Gini, Entropy
- ④ min_sample_leaf node : 20





✓ { min samples split = 50
min-sample leaf node = 35

$F_1 <$

Area > 5992

Actual
↑

3

10000, 20000, 12000 = 14000

5

Actual
↑

2

25000, 50000

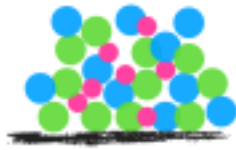
✓ A_1 10000 $P_1 = 14000$
✓ A_2 20000 $P_2 = 14000$
✓ A_3 12000 $P_3 = 14000$

A_1 25000

A_2 50000

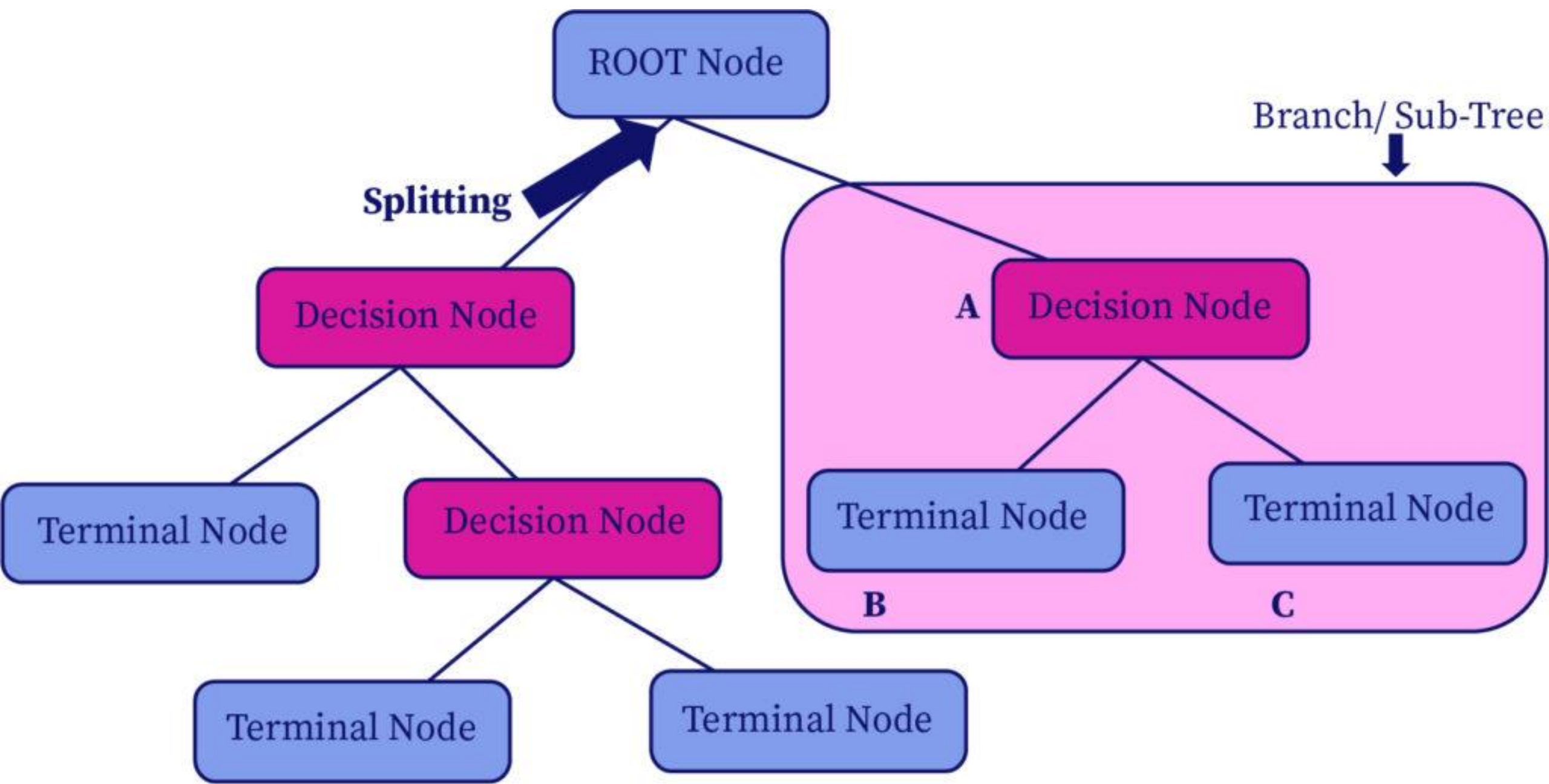
37500

37500



diameter size

colour



Note :- A is parent node of B and C.

